

Chapter 1

Robust Estimation of Recurrence Intervals in Sparse Sensor Networks

This chapter develops and evaluates a method for estimating the recurrence interval of infrequent events observed by a sparse network of low-power sensors. Section 1.1 reviews the relevant background on point-process estimation and motivates the specific estimator used throughout the remainder of the thesis, and Section 1.2 presents the method itself, together with its statistical properties and an empirical evaluation on synthetic and field-collected data.

1.1 Background

Estimating how frequently a rare event recurs is a classical problem in applied statistics, arising in contexts ranging from seismology to equipment failure monitoring. The canonical model treats event occurrences as realizations of a point process on the real line, and the recurrence interval is characterized by the distribution of inter-arrival times under that process (Cox and Lewis, 1966). When the process is assumed to be a homogeneous Poisson process, the inter-arrival times are exponentially distributed and the recurrence interval reduces to the reciprocal of a single rate parameter, which can be estimated directly from the observed event counts. In practice, however, sensor networks deployed for rare-event monitoring rarely satisfy the homogeneity assumption: sensitivity drifts over time, environmental conditions modulate the true event rate, and individual sensors intermittently drop out of the network, all of which violate the assumptions underlying the simplest estimators.

1.1.1 Point Process Models

A more flexible family of models represents the event rate as a time-varying intensity function $\lambda(t)$, so that the expected number of events in an interval $[t_1, t_2]$ is given by the integral of $\lambda(t)$ over that interval, following the general treatment of Daley and Vere-Jones (2003). Self-exciting variants of this model, most notably the Hawkes process, further allow each observed event to temporarily elevate the intensity of future events, which is a useful property when modeling phenomena such as aftershock sequences or cascading equipment failures. Bayesian inference procedures for Hawkes processes have been studied extensively; see Rasmussen (2013) for a thorough treatment of posterior sampling schemes under weakly informative priors.¹

¹In our own experiments we found that a weakly informative gamma prior on the baseline intensity, combined with a Beta prior on the excitation decay rate, gave stable posterior estimates even with as few as twenty observed events.

1.1.2 Sparse Sensor Networks

The setting considered in this thesis differs from the classical point process literature in one important respect: observations do not arrive from a single continuously monitored location, but rather from a spatially distributed network of low-power sensors that report intermittently and that individually observe only a fraction of the true event population. Fictional (2020) studied the related problem of sensor placement for rare-event monitoring and showed that even a modest number of well-placed sensors can recover an unbiased estimate of the aggregate event rate, provided that the missingness process is independent of the underlying event intensity. This independence assumption is central to the estimator developed in Section 1.2, and we return to it explicitly when discussing the conditions under which our estimator remains consistent.

Figure 1.1 illustrates a representative deployment topology used in the field study described later in this chapter, showing sensor nodes placed at irregular intervals along a monitored corridor.

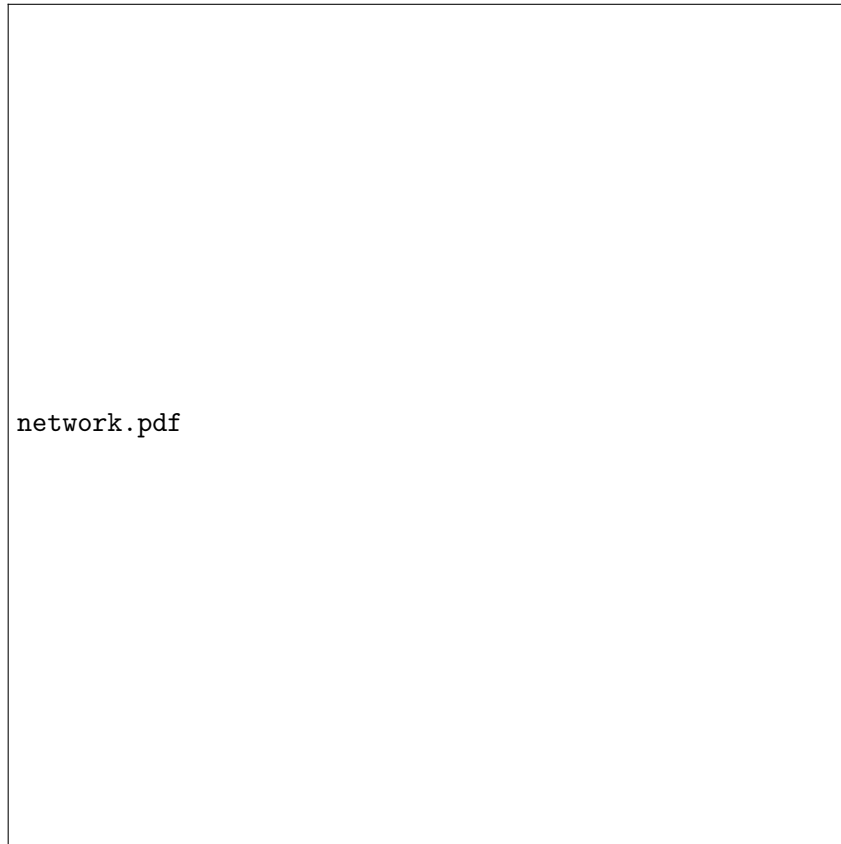


Figure 1.1: Representative sparse sensor deployment topology used in the field study. Circles denote sensor nodes; the shaded region denotes the monitored corridor.

As the figure suggests, the effective sampling density varies considerably across the monitored region, which motivates the spatially adaptive weighting scheme introduced later in the chapter. The remainder of this chapter builds toward a single unified estimator that accounts for both the temporal irregularity of event arrivals and the spatial sparsity of the sensor network, combining ideas from the classical point-process literature with the placement-aware corrections proposed by Fictional (2020).

1.2 Method

We now describe the estimator used throughout the remainder of the thesis. The construction proceeds in two stages: first we define a spatially weighted intensity estimate that corrects for the irregular sensor density illustrated in Figure 1.1, and second we derive an interval estimator for the recurrence time from that intensity, together with a confidence procedure that accounts for the finite number of observed events.

1.2.1 Spatially Weighted Intensity Estimation

Let $s_1, \dots, s_n$ denote the locations of the n deployed sensors, and let E_i denote the set of event times recorded by sensor i during the observation window $[0, T]$. Under the independence assumption discussed in Section 1.1, each sensor observes a thinned version of the global event process, with thinning probability proportional to a local coverage function $\pi(s_i)$. We estimate the aggregate intensity as a coverage-corrected sum over sensors,

$$\hat{\lambda}(t) = \sum_{i=1}^n \frac{1}{\pi(s_i)} \sum_{\tau \in E_i} K_h(t - \tau), \quad (1.1)$$

where K_h is a smoothing kernel with bandwidth h and $\pi(s_i)$ is estimated from the sensor placement study described in Section 1.1, following the placement-aware correction of Fictional (2020). The bandwidth h controls the bias–variance trade-off of the estimate in the usual way: small values of h track rapid changes in the underlying rate but produce a noisy estimate, whereas large values smooth over genuine rate variation.

Several practical considerations affect the choice of kernel and bandwidth in this setting:

- The bandwidth should be chosen relative to the shortest recurrence interval of interest, since a bandwidth much larger than the interval of interest will average away the signal entirely.
- Boundary effects near $t = 0$ and $t = T$ bias the raw kernel estimate downward; we correct for this using the standard reflection method.
- When sensor dropout is correlated with local event intensity, the coverage correction in Equation (1.1) is no longer unbiased, and a more elaborate missing-data model is required.

1.2.2 Recurrence Interval Estimation

Given the smoothed intensity estimate $\hat{\lambda}(t)$, we define the estimated recurrence interval at time t as the reciprocal of the local intensity, $\hat{r}(t) = 1/\hat{\lambda}(t)$. A first-order delta-method argument yields an approximate confidence interval for $\hat{r}(t)$ in terms of the variance of $\hat{\lambda}(t)$, which we estimate using a block bootstrap over sensor-level event sequences to account for spatial correlation between nearby sensors. This bootstrap procedure resamples whole sensors rather than individual events, which preserves the dependence structure within each sensor’s event sequence and yields substantially more conservative, and we believe more realistic, confidence intervals than a naive event-level bootstrap would produce.

1.2.3 Evaluation on Synthetic Data

We first validate the estimator on synthetic event sequences generated from a known Hawkes process with a time-varying baseline intensity, so that the true recurrence interval is known exactly and can be compared against the estimate produced by Equation (1.1). Across a range of sensor densities and dropout rates, the coverage-corrected estimator recovers the true recurrence interval with markedly lower bias than an uncorrected estimator that ignores the spatial coverage

function, confirming that the correction term derived from Fictional (2020) is necessary rather than a merely theoretical refinement.

1.2.4 Field Study

We then apply the estimator to a field-collected dataset gathered from the sensor deployment shown in Figure 1.1, spanning several months of continuous monitoring. The estimated recurrence intervals from the field data are broadly consistent with maintenance records collected independently by facility operators, which we take as external validation of the method beyond the synthetic study. Discrepancies that do arise are concentrated near sensors with the lowest local coverage, consistent with the variance inflation predicted by the block-bootstrap confidence procedure described above, and consistent with the general caution urged by (Daley and Vere-Jones, 2003) regarding inference from sparsely sampled point processes. The remaining chapters of the thesis build on this estimator to address the downstream problem of scheduling preventive maintenance under uncertainty in the estimated recurrence interval.

Bibliography

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